



Traffic Light Scheduling Optimization – Minimizing congestion at a 4- way intersection

Members: Steve George, Kaushal Nair
Instructor: Aida Sharif Rohani
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Why this problem matters?

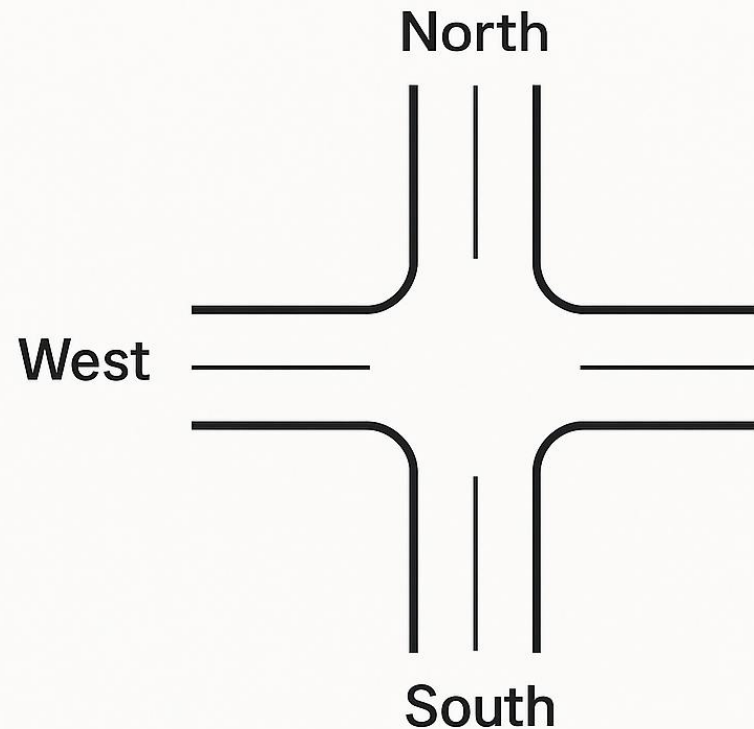
- ▶ Congestion costs
- ▶ Limitations of Fixed-Time Signals
- ▶ Economic Impact



Core Research Question

How can we use algorithmic decision-making to minimize queue lengths and wait times at a 4-way intersection?

Modeling an Intersection



TIMING CONSTRAINTS

-  Minimum Green: 10 seconds
-  Maximum Green: 60 seconds
-  Yellow Phase: 3 seconds
-  Departure rule: 1 vehicle per second during green
-  Arrival-model: deterministic steady-arrival simulation

Duration: 300 seconds (5 minutes)

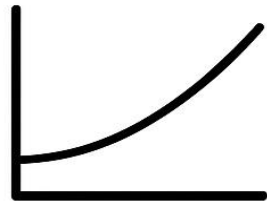
Three Competing Approaches

Greedy Algorithm



Prioritize lane with the longest queue

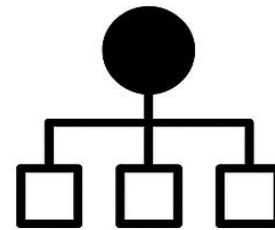
Sorting-Based Heuristic



$$\alpha = 1,0, \beta_s = 0,5$$

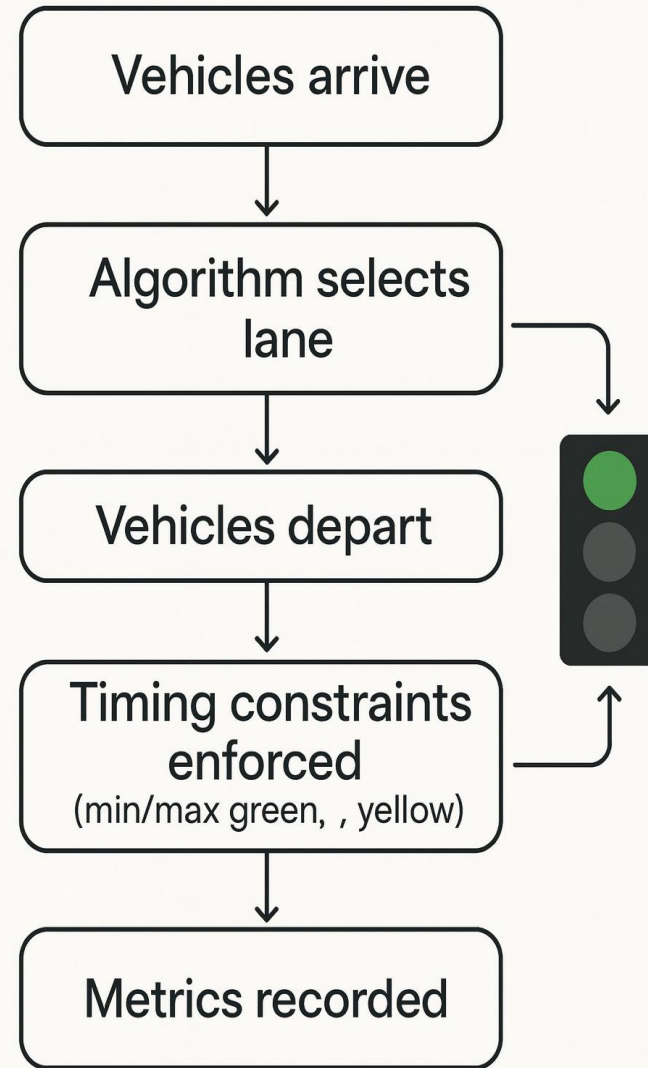
Function of Queue Length ($\alpha = 1,0$),
Arrival Rate ($\beta = 5$),
Fairness / Time Since Last Service ($\gamma = 0,3$)

Dynamic Programming



Simulates all feasible actions at each decision point

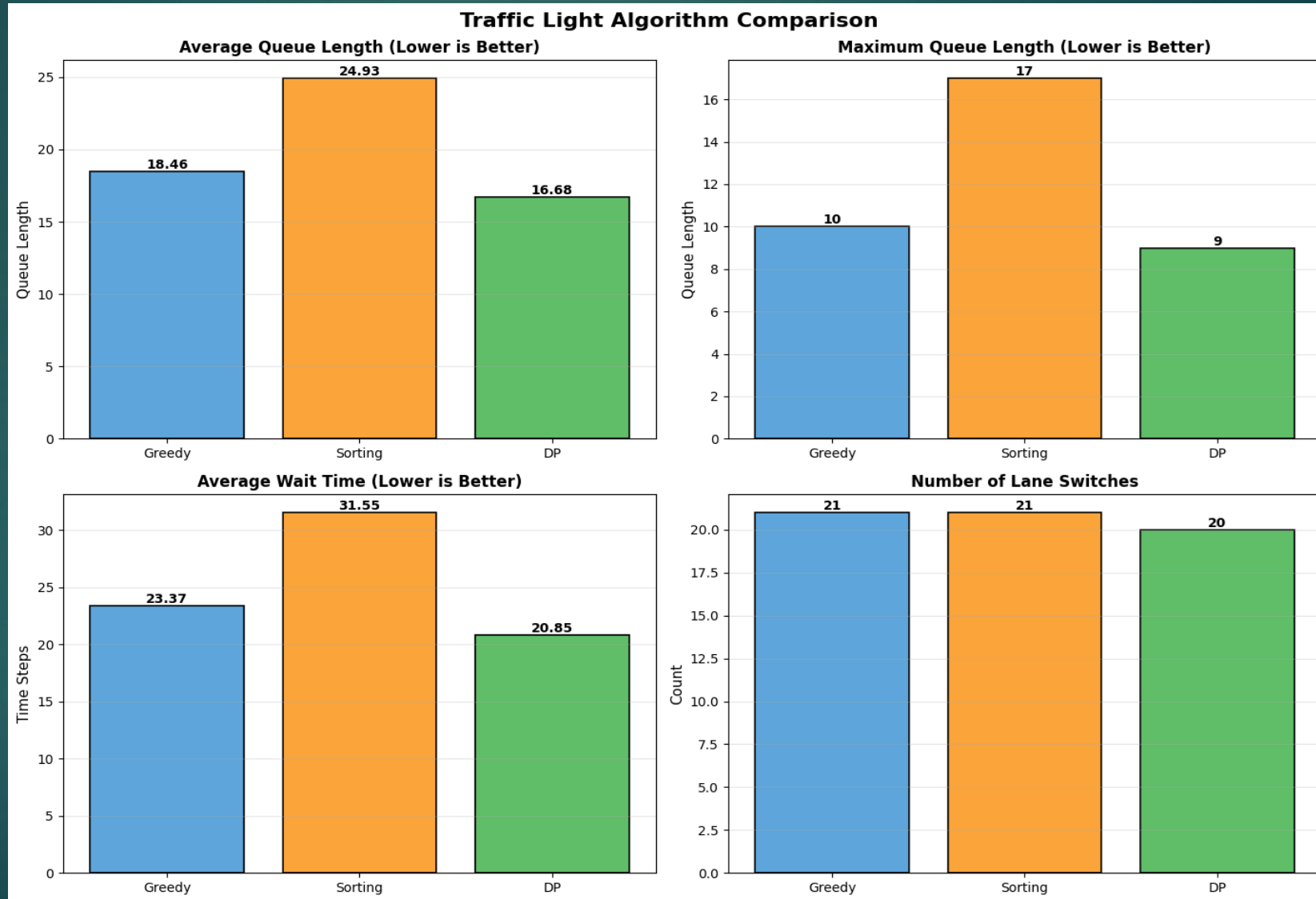
Simulation Framework

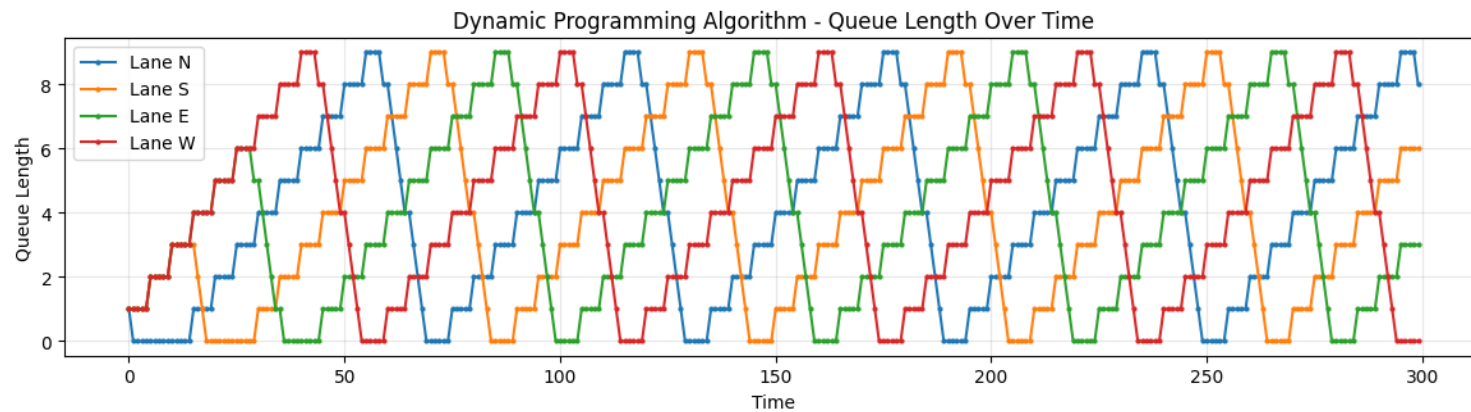
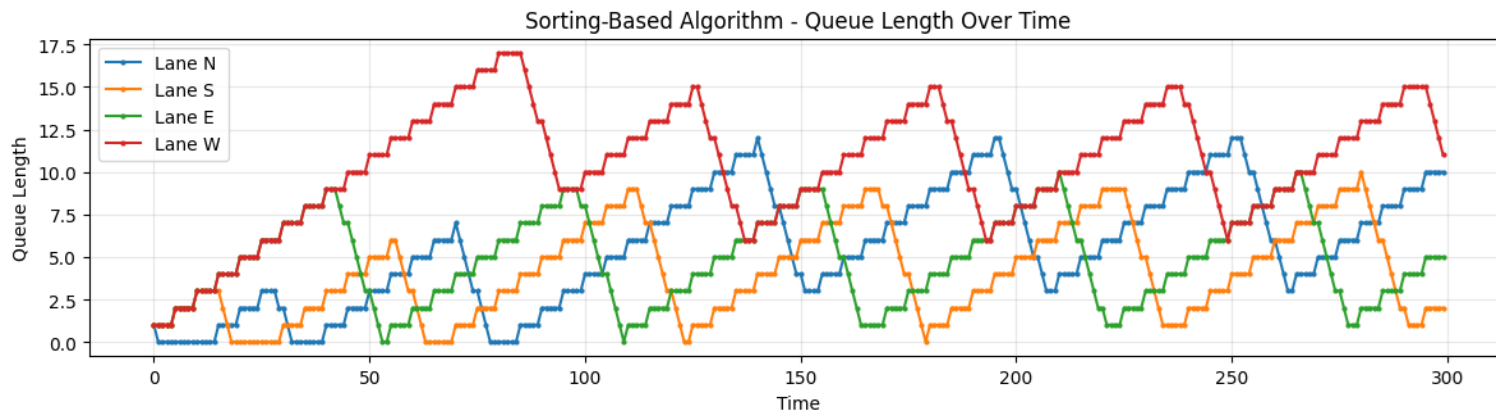
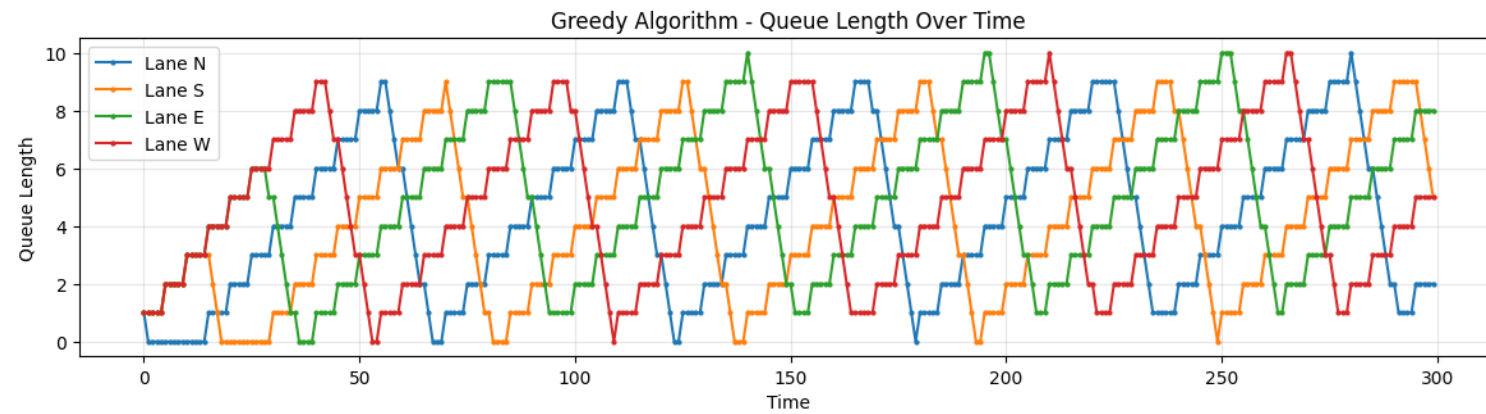


Performance Metrics

- ▶ Average Queue Length
- ▶ Maximum Queue Length
- ▶ Average Wait Time
- ▶ Lane Switches and Fairness

Results: Comparison Chart





Queue behavior over time



- ▶ Correct arrival timing
- ▶ Queue evolution
- ▶ Enforcement of minimum and maximum green times
- ▶ Correct yellow-phase transitions
- ▶ Accurate lane switching mechanics
- ▶ Algorithm decision validity
- ▶ Metrics correctness
- ▶ End-to-end system consistency

Limitations and Future Work

- ▶ Current models assume uniform traffic arrivals and analyze only single intersections without vehicle turns or pedestrian data.
- ▶ Dynamic Programming methods require high computational power, posing challenges for real-time traffic management.
- ▶ Future work aims to use real traffic data, multi-intersection networks, machine learning, and emergency vehicle prioritization.

Thank You!!!

